

Math515 – Financial Mathematics II

Spring 2014

Homework 3

1. Show that *almost surely* $\sum_{j=0}^{n-1} (\Delta W_j)^4 \rightarrow 0$ as the uniform time step size $\Delta t = h \rightarrow 0$ for a fixed final time $T = nh$. (Hint: use that $E[|Z|^4] = 3$ for any standard normal random variable Z .)
2. Use the Ito's formula to deduce the differential dX_t for the stochastic process $X_t = 2 + t + e^{W_t}$.
3. Use the Ito's formula to find a SDE satisfied by the stochastic process $X_t = \cos(t^2 + W_t)$.
4. Use the Ito's formula to show that the following are the solutions to the SDEs:
 - (a) $X_t = (1 + t)^2(X_0 + t + W_t)$ satisfies $dX_t = \left[\frac{2X_t}{1+t} + (1+t)^2\right] dt + (1+t)^2 dW_t$;
 - (b) $X_t = (1 + W_t)^2$ satisfies $dX_t = dt + 2\sqrt{X_t}dW_t$ with $X_0 = 1$;
 - (c) $X_t = \log(1 + W_t)$ satisfies $dX_t = -\frac{1}{2}e^{-2X_t}dt + e^{-X_t}dW_t$ with $X_0 = 0$.
5. Let $I_t^{(0)} = 1$, $I_t^{(1)} = W_t$, $I_t^{(2)} = W_t^2 - t$, $I_t^{(3)} = W_t^3 - 3tW_t$, and $I_t^{(4)} = W_t^4 - 6tW_t^2 + 3t^2$. Use the Ito's formula to show $dI_t^{(n)} = nI_t^{(n-1)}dW_t$ and then determine $I_t^{(5)}$ and $I_t^{(6)}$.
6. Use the Ito's formula for the following
 - (a) Since $X_t = t + W_t$ satisfies $dX_t = dt + dW_t$, confirm that $Y_t = \sinh X_t$ satisfies $dY_t = \left[\frac{1}{2}Y_t + \sqrt{1 + Y_t^2}\right]dt + \sqrt{1 + Y_t^2}dW_t$.
 - (b) For the above Ito process Y_t , deduce dZ_t for the Ito process $Z_t = Y_t^3$.